A Noise-Shaped Switching Power Supply Using a Delta–Sigma Modulator

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***Abstract*—A technique to reduce in-band tones in switch-mode power supplies is described. It takes advantage of the noise-shaping properties of the delta–sigma ($\Delta\Sigma$) modulator to eliminate the spikes normally present in switching power supplies. A framework is introduced for comparing the conventional pulsewidth modulated (PWM) controller and this approach. A buck converter test circuit is constructed that is designed for a PWM controller clocked at 200 kHz and then substituted with a $\Delta\Sigma$ modulator controller clocked at 400 kHz. The rms noise power of the PWM controller is 14.9 mW compared to the rms noise power for the $\Delta\Sigma$ modulator of 75.85 mW measured in a 2-MHz bandwidth. Although the $\Delta\Sigma$ modulator rms noise power is higher, the noise floor is below the tones seen at the output of the PWM controller. A multibit $\Delta\Sigma$ modulator controller, however, provides a significant reduction in the spectral output of the power supply. Values of 3.75 and 0.24 mW rms noise power are observed at the output of a 2-bit and 4-bit $\Delta\Sigma$ modulator controller, respectively.**

***Index Terms*—Delta–sigma ($\Delta\Sigma$) modulator, pulsewidth modulation (PWM).**

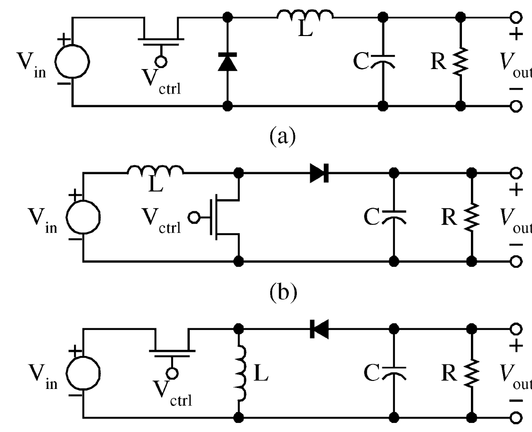

Fig. 1. Three fundamental types of SMPSs. (a) Buck. (b) Boost. (c) Buck-boost.

I. Introduction

POWER supplies are fundamental components in all electronic equipment. It is challenging to develop economical power supplies with single or multiple dc outputs from a single dc source that are regulated and spectrally quiet. Switching techniques such as in switch-mode power supplies (SMPSs) are a common method for producing regulated voltages. These power supplies offer efficient power conversion—often greater than 90% and reliable output voltages even in the presence of changing loads or varying input voltages [1].

The three main types of switching power supplies are shown in Fig. 1 and are the buck, boost, and buck-boost power supplies. These devices work by switching, or modulating, the input so that a current is developed through an inductor. This, in turn, injects charge onto a capacitor that holds the output voltage constant. All three types of SMPSs regulate the output, even in the presence of changing source voltages and/or loading impedances. This regulation is achieved through a feedback loop that controls the modulation of the switch input.

The conventional method of implementing the modulation control signal is a technique known as pulsewidth modulation (PWM). A PWM controller uses a fixed frequency varying duty cycle modulation scheme to achieve the desired control signal. One drawback of the PWM approach is that ripple exists in the output voltage due to clock feedthrough. This ripple appears as a noise source added to the desired dc value. The PWM noise is manifested as impulses, or tones, at integer multiples of the clock frequency and is undesirable because it is coupled into all the components that receive power from the SMPS.

This paper presents an alternative control solution that uses a delta–sigma ($\Delta\Sigma$) modulator in place of the PWM controller. This new control method, while increasing the overall noise at the output, shapes the noise and reduces the tones that exist in the PWM controller [2]. In addition to presenting this approach, analysis techniques are developed to compare the PWM and $\Delta\Sigma$ modulator controllers. The $\Delta\Sigma$ modulator controller is suitable for applications sensitive to high frequency noise and tonal behavior such as in RF designs or communication applications. Section II describes the PWM and $\Delta\Sigma$ modulator-based controllers and characteristics of their noise spectrums. Section III provides a framework for comparing the efficiency, stability, and spectral performance of the controllers. This is illustrated through an experimental example in Section IV. Section V presents other possible configurations of the $\Delta\Sigma$ modulator controllers including a multibit implementation. Finally, the work is summarized in the conclusion.

II. $\Delta\Sigma$ Modulator-Based SMPS Controller

Before comparisons of the PWM and $\Delta\Sigma$ modulator-based controllers are given, a brief background of each is provided. For the discussion that follows, a buck-type SMPS is selected. It is expected that the typical characteristics of the PWM and $\Delta\Sigma$ modulator-based controllers will be similar with any one of the controllers.

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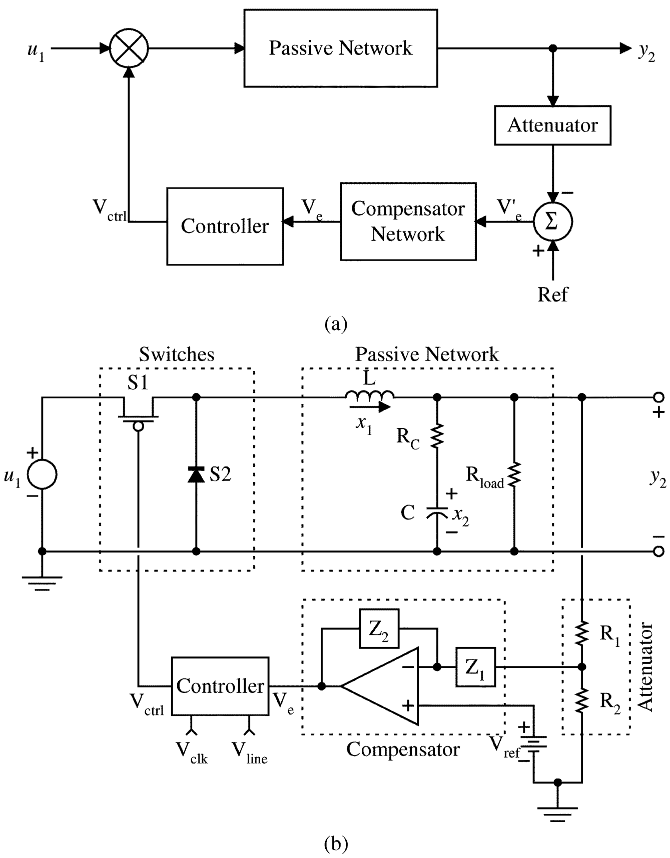

Fig. 2. Typical buck-type SMPS. (a) Block diagram. (b) Circuit diagram.

Fig. 2(a) is a typical block diagram of a buck-type switching power supply. A typical circuit of each of the blocks is shown in the schematic of Fig. 2(b). The inputs to the switching power supply are u_i, the outputs are y_i, and the state variables are x_i. These variables will be used later in the analysis of the controllers. All other component values are self-explanatory with the possible exception of R_C which represents the equivalent series resistance (ESR) of the capacitor that can have a significant effect on the transfer function of the SMPS, as explained later.

As seen in Fig. 2(a), the control signal V_{ctrl} is multiplied by the input u_1. In this case, the input has been introduced as a variable; however, the analysis can be simplified by assuming that the input does not change. This results in scaling of the input which can also be represented by a gain stage in the flow diagram. The control signal that has been scaled by the input $u_1 \cdot V_{\text{ctrl}}$ is then passed to the passive low-pass filter (LPF) network. The dc and low-frequency components are passed in the filter while all other ac components are attenuated. Once the output is generated, it is observed by the feedback loop. The feedback loop is comprised of three main components: a differencing mechanism that subtracts the output of the power supply from a temperature compensated stable reference such as a bandgap reference V_{ref}; a compensation network that is an active filter used to contribute gain and designed to stabilize the closed loop; and a controller that is a nonlinear device used to convert the error value that has been filtered by the compensation device into a time-varying modulation signal. The modulation signal is then used to control the power-supply switches, and therefore, closes the feedback loop. The controller block has been left arbitrary here to suggest the placement of our different types of control structures. The two additional inputs to the controller V_{line} and V_{clk} determine the gain and operating frequency of the controller.

A. PWM Controller

A PWM controller is shown in Fig. 3(a). The PWM controller is a comparator that compares the incoming error voltage to a sawtooth. This is illustrated by the waveforms in Fig. 3(b). When the sawtooth crosses the input signal (top of figure), the output changes state (bottom of figure). Also shown in Fig. 3(a) is a simple sawtooth waveform generator built out of a resistor, a capacitor, and a transistor which uses a *RC* charging waveform to approximate a sawtooth wave. This sawtooth waveform generator is driven by a pulsed clock signal V_{clk}, with a period equal to the desired operating frequency. The other input voltage V_{line} is normally another reference voltage but can also be derived from the input voltage u_1. Deriving V_{line} from the input allows for feedforward control in addition to the feedback control already present. While this feedforward mechanism is not used in the calculations that follow, it should be noted that this does allow for more efficient control, or line regulation, of the input. The output of the comparator is then delivered to a driver circuit that normally adds current gain to the signal to assist in switching the power transistor more quickly—giving the power transistor sharper transitions. While the value of V_{ctrl} does take on analog voltage values, the calculations are simplified by representing it as a digital signal normalized to the two values of 0 and 1.

The PWM controller uses a fixed-frequency varying duty cycle modulation scheme to achieve the desired control signal where a value, d, is introduced to describe the percentage of on time of the PWM waveform. This knowledge can be used to analyze the power-spectral density (PSD) of the PWM controlled power supply. First, define the term d as

$$d = D + \hat{d} \tag{2.1}$$

where D is the steady-state or nontime varying value of d and $\hat{d}$ is the small-signal perturbation about D used to compensate for varying input and/or loading. Then,

$$y_{20} = u_{10} \cdot D \tag{2.2}$$

where u_{10} is the steady-state value of the input u_1 and y_{20} is the steady-state output value y_2, where both of these are dc or nontime varying signals. The value for V_{ctrl} is now defined as

$$V_{\text{ctrl}} = \begin{cases} 1, & (n)T \le t < (n+d)T \\ 0, & (n+d)T \le t < (n+1)T \end{cases} \tag{2.3}$$

and

$$V_{\text{ctrl}}^{\text{SS}} = \begin{cases} 1, & (n)T \le t < (n+D)T \\ 0, & (n+D)T \le t < (n+1)T \end{cases} \tag{2.4}$$

where $V_{\text{ctrl}}^{\text{SS}}$ is the steady-state value of V_{ctrl}. Next, the Fourier transform of $V_{\text{ctrl}}^{\text{SS}}$ is found

$$F\left\{V_{\text{ctrl}}^{\text{SS}}\right\} = 2\pi D \cdot \operatorname{sinc}\left(\frac{\omega DT}{2}\right) \cdot \sum_{n=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi n}{T}\right) \cdot e^{-j\left(\frac{\omega DT}{2}\right)} \tag{2.5}$$

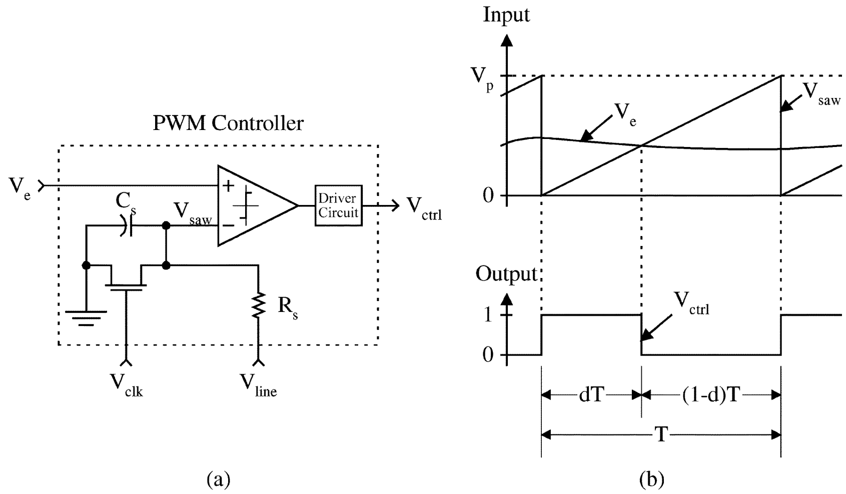

Fig. 3. PWM controller. (a) Circuit diagram. (b) Input and output waveforms.

where ω is the independent variable for frequency measured in radians per second. Assuming a unit load output impedance, i.e., $Z_{\text{out}} = 1$, the PSD for this function is

$$S_{V_{\text{ctrl}}^{\text{SS}}} = \frac{\left|F\left\{V_{\text{ctrl}}^{\text{SS}}\right\}\right|^2}{\text{Re}\{Z_{\text{out}}\}} = (2\pi D)^2 \cdot \text{sinc}^2\left(\frac{\omega DT}{2}\right) \cdot \sum_{n=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi n}{T}\right) \qquad (2.6)$$

where $S_{V_{\text{ctrl}}^{\text{SS}}}$ is the PSD of $V_{\text{ctrl}}^{\text{SS}}$. As seen in (2.6), the PSD for the PWM controller is a series of impulses that are scaled by a sinc function. These impulses occur at integer multiples of the clock frequency and they are sometimes referred to as tones. This signal is scaled by the input, u_1 and fed into the passive LPF where the ac terms are attenuated. The LPF attenuation, however, does not eliminate the tones completely. The clock signal and its first harmonic receive the least attenuation and are the major contributors to clock feedthrough noise seen at the output. The output of the PWM controlled SMPS clocked at 200 kHz, taken after the passive LPF, is shown in Fig. 4(a), the dc term, or desired output, has been subtracted out of this graph to show only the noise spectra. The data for this graph was generated in HSPICE using a test circuit that will be described in Section IV.

To reduce the clock feedthrough noise, an additional LPF can be introduced at the output of the power supply that is not contained within the control loop. Added filtering, however, increases the part count and overall cost of the power supply. One solution to reducing the clock feedthrough noise has been suggested in [2] where a dither signal is added to the clock controlling the PWM. This effectively adds random noise to the transition point in the PWM output. This scheme smears the tones into a Gaussian profile but does not eliminate them.

B. $\Delta\Sigma$ Modulator-Based Controller

The $\Delta\Sigma$ modulator produces a fixed period varying-frequency pulse waveform known as pulsecode modulation (PCM). It is a variation of the burst-mode controller because

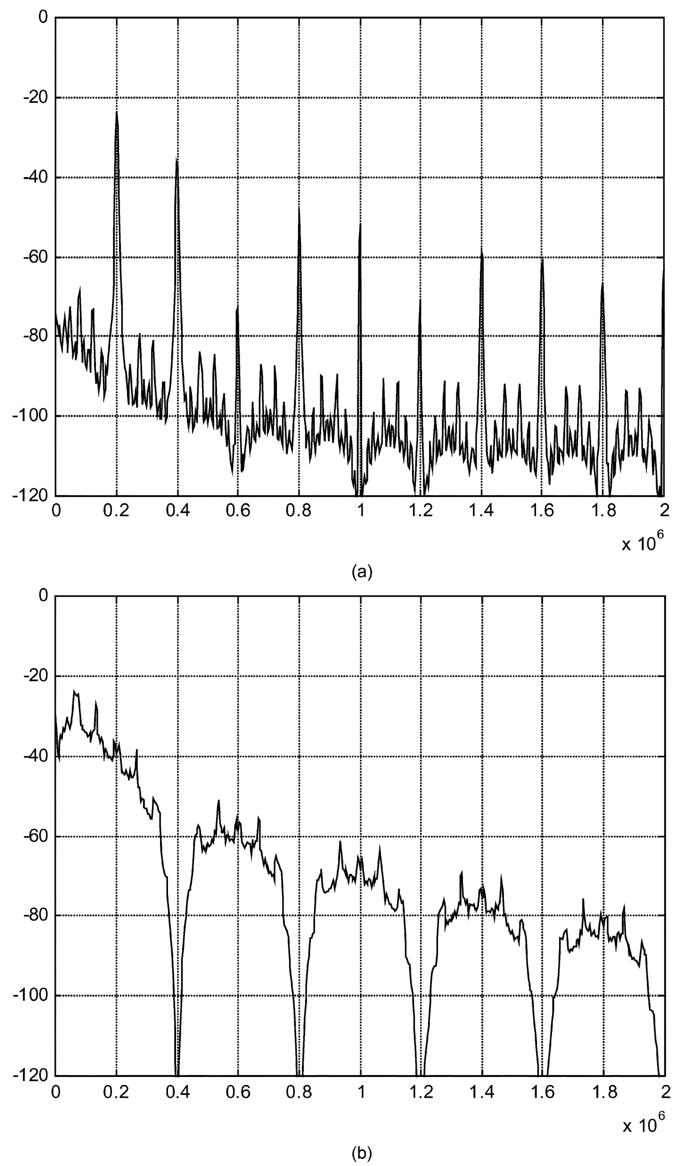

Fig. 4. (a) PSD of the PWM controlled SMPS clocked at 200 kHz. (b) PSD of the $\Delta\Sigma$ modulator controlled SMPS clocked at 400 kHz.

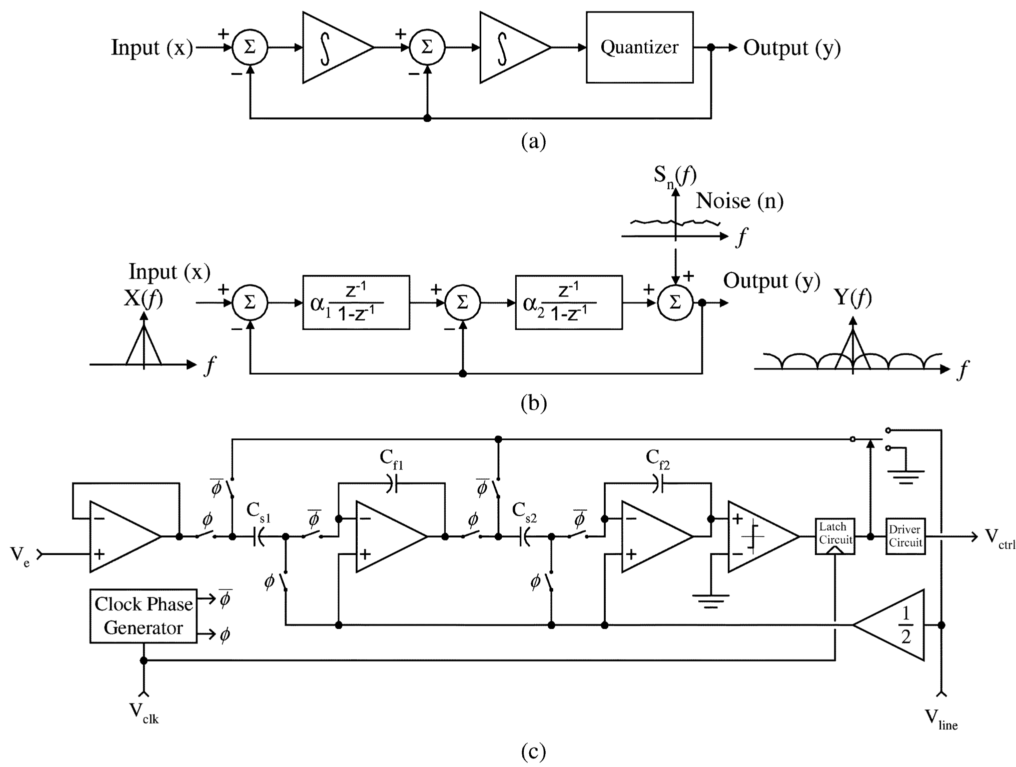

Fig. 5. (a) Block diagram for the second-order $\Delta\Sigma$ modulator. (b) Linearized z-domain model of second-order $\Delta\Sigma$ modulator. (c) Switched-capacitor $\Delta\Sigma$ modulator.

of its fixed period output [3]. A second-order $\Delta\Sigma$ modulator consists of two integrators followed by a comparator as shown in Fig. 5(a). The input and the feedback value of the output are subtracted at the front-end of the modulator to produce an error signal. As the output signal begins to track the input signal, the error signal approaches zero. The $\Delta\Sigma$ modulator reduces subharmonic tones by shaping the quantization noise and effectively randomizing the input to the comparator. As the order of the $\Delta\Sigma$ modulator is increased, the susceptibility to limit cycles (or subharmonic tones) is reduced [4].

To illustrate the noise-shaping properties of the $\Delta\Sigma$ modulator, a linearized z-domain model is used as shown in Fig. 5(b). The comparator quantization noise is modeled as a white noise source. The output is defined by the following equation:

$$Y = Xz^{-2} + N(1 - z^{-2}) \tag{2.7}$$

where X is the input signal and N is the additive quantization noise. The input sees only a linear delay passing through the $\Delta\Sigma$ modulator while the quantization noise is high pass filtered. Since the $\Delta\Sigma$ modulator is a sampled-data device, the filter for the noise is aliased in frequency at integer multiples of one-half the clock frequency. This gives the $\Delta\Sigma$ modulator output a comb filter appearance when viewed at multiples of the clock frequency. If a digital low-pass decimation filter follows the modulator, a significant portion of the high frequency noise is eliminated. Unfortunately, adding a decimation filter after the modulator introduces a relatively large latency. However, when placed in the SMPS, the power-supply components perform this filtering and thus, eliminate the latency and the requirement for additional circuitry.

The $\Delta\Sigma$ modulator can be directly substituted for the PWM as the controller of Fig. 2(b). The advantage of using the $\Delta\Sigma$ modulator controller is clear once the output spectrum of the power supply is evaluated. When the $\Delta\Sigma$ modulator is placed in the feedback portion of the switching power-supply control loop, the passive elements of the power supply act as a LPF. Consequently, the power-supply filters the high-frequency noise generated by the modulator as shown in Fig. 4(b). In this plot, a second-order $\Delta\Sigma$ modulator clocked at 400 kHz is substituted for the PWM controlled SMPS that was used to generate the graph in Fig. 4(a). HSPICE simulations with a switched-capacitor second-order $\Delta\Sigma$ modulator are used [9]. Notice the nulls occurring at the clock frequency. Also, visible are limit-cycle spikes at one-sixth of the clock frequency that have not been fully eliminated by the $\Delta\Sigma$ modulator. Adding dither will significantly reduce these spikes.

III. Efficiency and Stability Analysis of the PWM and $\Delta\Sigma$ Modulator Controllers

In order to compare the SMPS output noise specturm generated by the PWM and $\Delta\Sigma$ modulator controllers, equivalent designs must be generated. To make equivalent designs, the power transfer efficiency and stability must be examined. In this section, the efficiency of the SMPS is first analyzed. The differences between the PWM and $\Delta\Sigma$ modulator controller efficiency mechanisms will be explored.

A. *Efficiency in a SMPS*

In order to compare the PWM SMPS to the $\Delta\Sigma$ SMPS, we must first verify—or enforce by design—equivalent efficiencies

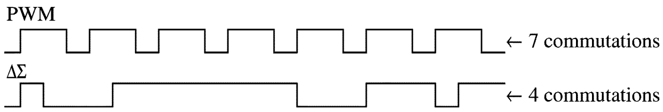

Fig. 6. PWM and $\Delta\Sigma$ modulator waveforms.

in both systems. To accomplish this, we begin with the definition of efficiency

$$\eta = \frac{P_o}{P_i} \tag{3.1}$$

where P_o is the output power, P_i is the input power, and η is the efficiency. We can further define P_i to be the sum total of the output power P_o and losses due to static and dynamic behavior of the system P_s and P_d, respectively

$$\eta = \frac{P_o}{P_o + P_s + P_d}. \tag{3.2}$$

For both systems to be equivalent, P_o must be the same for equivalent loads. Likewise, losses due to static elements, e.g., parasitic resistances will be the same because of the fact that both systems must be in the ON–OFF states an equivalent amount of time to produce equivalent outputs. The worst case ac losses [5] are

$$P_d = u_1 I_0 \frac{T_s}{T_{\mathrm{clk}}} \tag{3.3}$$

where u_1 is the input voltage, I_o is the average inductor current, T_s is the transition time when switching from low to high or high to low. This assumes that $t_{\mathrm{rise}} = t_{\mathrm{fall}} = T_s$ and T_{clk} is the clock period.

The dynamic losses for each system are directly related to the switching of the respective systems. This means that as the switching frequency is increased, the dynamic losses increase, or the efficiency decreases. As shown in Fig. 6, the average amount of switching that occurs in the $\Delta\Sigma$ modulator output for a dc input is less than that of the PWM. There are two reasons for this. First, the $\Delta\Sigma$ modulator acquires and holds an output value for an entire clock cycle, effectively halving the clock. Second, because the $\Delta\Sigma$ modulator output has some randomness, similar valued outputs can be bunched together. The PWM switching scheme is deterministic as compared to the $\Delta\Sigma$ modulator's statistical nature. Thus, the PWM will switch, or commutate, once per clock cycle. To make the two efficiencies equal, a proportionality constant k is introduced that creates the following relationship:

$$f_{\mathrm{clk}}^{(\Delta\Sigma)} = k f_{\mathrm{clk}}^{(\mathrm{PWM})}. \tag{3.4}$$

To determine the value of k, the statistical nature of the $\Delta\Sigma$ modulator is examined. Shown in Fig. 5(b) is the block diagram for a second-order $\Delta\Sigma$ modulator. The value for $x(t)$ at the input of the $\Delta\Sigma$ modulator is assumed to be a random variable with an average value μ_X, which is constant since a dc input is applied to the $\Delta\Sigma$ modulator. The input is assumed to be noisy given the fact that the input to the $\Delta\Sigma$ modulator is created by the closed loop operation of the SMPS. The output varies by the output tolerance value over time. Additionally, the probability of acquiring a truly dc value for $x(t)$ is zero from a real system that must contain some noise [6]. This is because the probability of achieving a range of values goes to zero as the bin width of the range decreases. Based on these facts, the output, $y(t)$, of the $\Delta\Sigma$ modulator also becomes a random variable. Thus, because of the feedback loops and memory inherit in the system, $y(t)$ can be described as a Markov sequence with two states A_L and A_H, the low and high amplitude, respectively [7]. The random variable $y(t)$ is a function of the order of the $\Delta\Sigma$ modulator and the random variable $x(t)$; however, $x(t)$ is ill defined. Even though the mean μ_X is known, the distribution, variance, or any other statistical moments are unknown. In SMPS systems with $\Delta\Sigma$ modulator controllers, the variance depends on the noise inherent in the circuit elements, the temperature, and/or the loading of the power supply. This suggests that the variance is time varying. Nevertheless, knowing that the output, $y(t)$, is a Markov sequence can help by providing a lower bound on the proportionality constant, k.

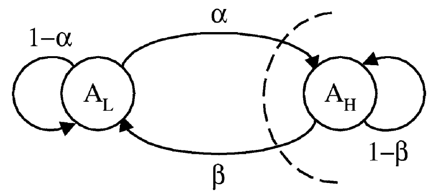

Fig. 7. Markov sequence state diagram for Y_n.

Fig. 7 shows the state diagram for the output of the $\Delta\Sigma$ modulator $y(t)$. The low and high output voltage amplitudes are A_L and A_H, respectively. The values listed on the transition lines α and β are the probabilities of transitioning from one state to the other. There are also stationary probabilities associated with each state μ_L and μ_H that are the probabilities of being in state A_L and A_H, respectively. The dashed line on the graph indicates a cut-set where the flow into the cut-set must sum to zero

$$\begin{aligned} \mu_L\alpha - \mu_H\beta &= 0 \\ \Rightarrow \mu_L\alpha &= \mu_H\beta \end{aligned} \tag{3.5}$$

This equation indicates that the probability of being in a particular state and moving it to the other is equal to μ_L or μ_H times the probability of transitioning α or β, respectively. These are the values of interest and they are inversely related to the scaling value, k

$$k = \frac{1}{\mu_L\alpha} = \frac{1}{\mu_H\beta} \tag{3.6}$$

and, since

$$0 \le \alpha, \beta, \mu_L, \qquad \mu_H \le 1 \tag{3.7}$$

then

$$k > 1.$$

As a result, the $\Delta\Sigma$ modulator can be overclocked by k and still maintain an efficiency equivalent to the PWM controller. The values of μ_L and μ_H are defined by the dc value of the input,

μ_X. Thus, by letting the input to the $\Delta\Sigma$ modulator average to a value V_i and also setting $A_L = 0$ and $A_H = V_p$, we obtain

$$\mu_H = \frac{\mu_X}{\mathrm{A_H} - \mathrm{A_L}} = \frac{V_i}{V_p}$$
$$\mu_L = 1 - \frac{\mu_X}{\mathrm{A_H} - \mathrm{A_L}} = 1 - \frac{V_i}{V_p} = 1 - \mu_H. \tag{3.8}$$

The values of α and β remain unknown because they depend on other factors not defined. By rearranging (3.6) and substituting for the values of μ_L and μ_H, the relationship between α, β and μ_H is

$$\alpha(1 - \mu_H) = \beta\mu_H$$
$$\alpha = \frac{\mu_H}{(1 - \mu_H)}\beta \tag{3.9}$$

and noting that if μ_H is fixed, k will become smaller as α and/or β become larger. The smallest value of k is found based on the given value of μ_H. The following equations will accomplish this using μ_H :

$$\alpha = \begin{cases} \frac{\mu_H}{(1-\mu_H)}, & \text{for } \mu_H \le \frac{1}{2} \\ 1, & \text{for } \mu_H > \frac{1}{2} \end{cases}$$
$$\beta = \begin{cases} 1, & \text{for } \mu_H \le \frac{1}{2} \\ \frac{(1-\mu_H)}{\mu_H}, & \text{for } \mu_H > \frac{1}{2}. \end{cases} \tag{3.10}$$

This ensures the maximum probability of transitioning between states, or the smallest value of k. For example, if α or β equal 1 when the corresponding state is reached, a state transition occurs on the next clock cycle. This establishes the lower bound, or worst case, for the scaling value k. We demonstrate this with the following example:

Let $A_L = 0 \quad A_H = V_p = 1$ and $V_i = \frac{1}{4}$

then $\mu_H = \frac{V_i}{V_p} = \frac{1}{4}$ which is $\le \frac{1}{2}$

so $\beta = 1$

and $\alpha = \frac{\mu_H}{(1 - \mu_H)} = \frac{1}{4}\frac{4}{3} = \frac{1}{3}$.

The state diagram is represented by Fig. 8. Thus, the overclocking value will be

$$k = \frac{1}{\mu_L \alpha} = \frac{1}{\frac{3}{4}\frac{1}{3}} = \frac{1}{\mu_H \beta} = \frac{1}{\frac{1}{4}\frac{1}{1}} = 4. \tag{3.11}$$

A means of calculating the overclocking value for the $\Delta\Sigma$ modulator relative to the PWM is now established for the SMPS. From (3.6) and (3.10), the absolute minimum value is determined for k as 2 when $\alpha = \beta = 1$ for a $\mu_H = (1/2)$. Note that the $\Delta\Sigma$ modulator holds an output for an entire clock cycle. Thus, for both the PWM and $\Delta\Sigma$ modulator systems to produce a square wave with a 50% duty cycle having the same frequency, the $\Delta\Sigma$ modulator must be clocked at twice the frequency of the PWM. Additionally, the efficiency of the $\Delta\Sigma$ modulator SMPS is dependent on the fractional value of μ_H that we wish to achieve, whereas the efficiency is fixed for the PWM. Using this reasoning, a value of 2 can be chosen for k which insures that the same or better efficiency as the PWM is achieved. If the input/output ratio will not have large excursions from the mean, increased values of k can be used.

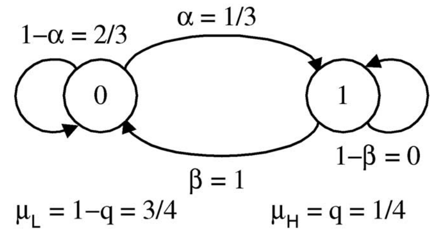

Fig. 8. Demonstration of the Markov sequence.

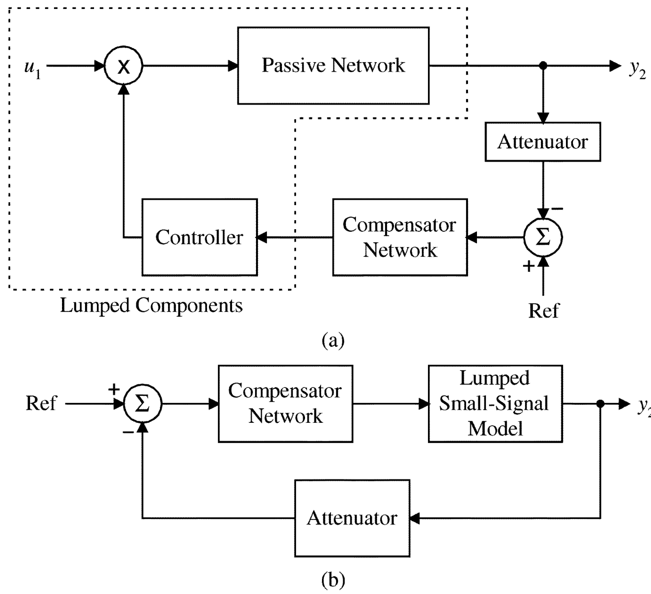

Fig. 9. Small-signal model. (a) Lumped components for state–space averaging. (b) Resulting block diagram.

B. Stability of the SMPS

There are many methods of determining the stability of a system. One common method used in SMPS designs is to plot the magnitude and phase response of the open loop system and calculate the phase margin associated with the system [1]. To use this approach, the SMPS system must first be linearized to allow analysis of the system with switch and comparator nonlinearities. To linearize the comparators and switches in the PWM and DS modulator controllers, a method of state–space averaging and linearization can be used that creates a small-signal model [1].

Fig. 9(a) shows which components are to be incorporated into the small-signal model and Fig. 9(b) shows the resulting block diagram with the small-signal model incorporated. Notice that the input variables have all been absorbed into the small-signal model. The input variables for the SMPS do not lend themselves naturally to normal control methods. Normally the input values, or independent variables, are added to the system by use of a summation block. In the case of the SMPS, the input voltage is multiplied by a modulated feedback signal. The loading resistance, which can vary, is used to describe the second input variable and is part of the passive LPF, and consequently, its variations affect the location of the poles of the passive LPF. Nevertheless, these values can be incorporated into the transfer

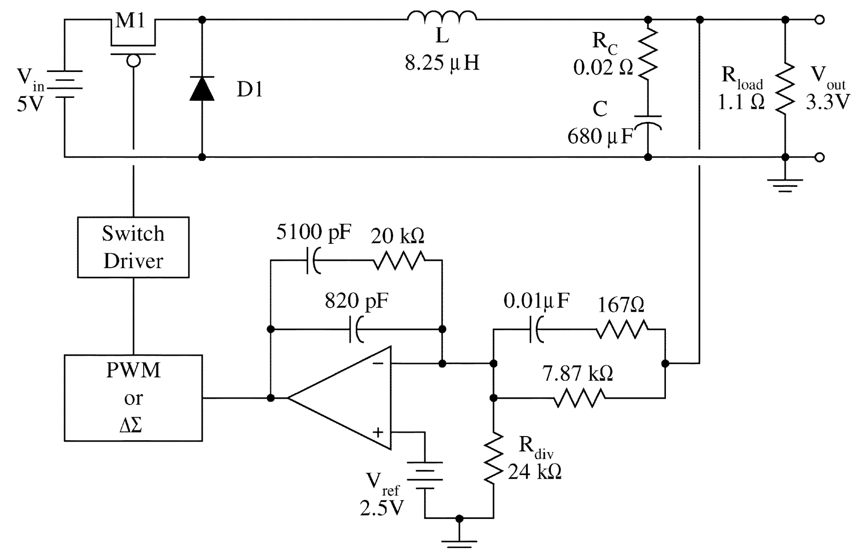

Fig. 10. SMPS circuit with component values used for validation.

function. However, the dynamic behavior is determined by performing a point by point analysis of the system for each different input.

For the PWM, there are two criteria that must be met to use state–space averaging: 1) the sampling rate of the controller must be large in comparison to the observed signal y_2, and 2) there must be some means of storing information from one discrete event to the next [1]. Both of these conditions are easily met in the PWM and $\Delta\Sigma$ modulator controllers. The PWM network samples much faster than the cutoff frequency of the system and the inductor and capacitor act as the memory storage elements. For linearization, we assume that the AC variations about the dc operating point are small. With these assumptions, state–space averaging and linearization as outlined in [1] yields the following transfer function:

$$G(s)H(s) = \frac{\frac{u_{10}K(s)R_C}{V_pL}\left(s+\frac{1}{R_CC}\right)}{s^2+s\left(\frac{1}{R_{\text{load}}}+\frac{R_C}{L}\right)+\frac{1}{LC}}$$

where

$$u_1 = \underbrace{u_{10}}_{\text{dcterm}} + \underbrace{\hat{u}}_{\substack{\text{small}\\\text{signal}\\\text{term}}} \tag{3.12}$$

where $G(s)H(s)$ is the open-loop transfer function and the components are shown in the schematic of Fig. 3(b). $K(s)$ is the transfer function of the compensation network, V_p is the peak value of the PWM sawtooth or the high-value feedback value A_H in the $\Delta\Sigma$ modulator.

For the $\Delta\Sigma$ modulator, there are three added difficulties in the analysis. The $\Delta\Sigma$ modulator has a finite delay associated with it because it is a discrete-time circuit when implemented with switched-capacitor techniques. Second, because the $\Delta\Sigma$ modulator is integrating, it acts like an LPF, and third, the $\Delta\Sigma$ modulator only switches at discrete-time intervals. The finite time delay mentioned first can be neglected if the oversampling ratio is large enough, i.e., the clocking frequency of the $\Delta\Sigma$ modulator is much higher then the crossover frequency of the SMPS, or there is suitable phase margin to compensate for the delay. However, to include this value, the delay is initially neglected in the analysis and then a value of e^{-sT} is used in the phase margin calculation to account for the delay. This adds $360\cdot f_T T$ degrees to the phase margin, where f_T is the crossover frequency of the transfer function and T is the sampling delay in the $\Delta\Sigma$ modulator. By forcing the passband frequency of the $\Delta\Sigma$ modulator to be much larger than the cutoff frequency of the SMPS, the LPF nature of the $\Delta\Sigma$ modulator can also be neglected. This is easily accomplished if the oversampling ratio is large. Finally, the issue of discrete-time switching can also be neglected for stability analysis if the sampling rate is high enough. This will ensure that the average of the input is mirrored to the output. The overclocking described above further helps validate this approach. Additionally, the discrete-time switching is what adds noise to the circuit. So, in general we can neglect all of the additional difficulties introduced by the $\Delta\Sigma$ modulator assuming that the switching frequency of the $\Delta\Sigma$ modulator is sufficiently large.

IV. Comparison of PWM and $\Delta\Sigma$ Modulator SMPS Controller

A specific example of the PWM and $\Delta\Sigma$ modulator controllers will be used to compare the two controllers. The SMPS used in this example is shown in Fig. 10 [8]. This SMPS is designed to convert a 5-V dc source voltage to 3.3 V with an output voltage tolerance of 10 mV at a 200 kHz operating frequency while supplying up to 10 W of power. Using the component values given in Fig. 10, the open-loop transfer function as described in (3.12) is calculated. It produces the Bode plots shown in Fig. 11. The resulting phase margin is approximately 60° at a crossover frequency of 40 krad/s, which is a sufficient value for stable operation.

A second-order $\Delta\Sigma$ modulator, as shown Fig. 5(c), was designed and substituted for the PWM controller of Fig. 10. The

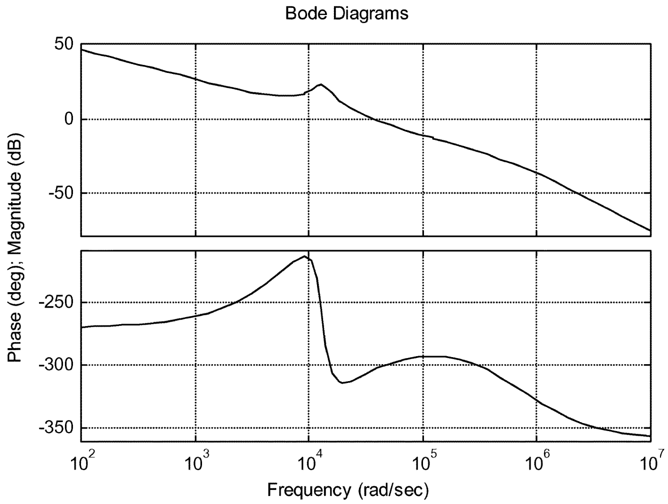

Fig. 11. Bode plots for the SMPS open-loop transfer function.

worst case overclocking value of $k = 2$ or $f_{\mathrm{clk}}^{\Delta\Sigma} = 400$ kHz and a crossover frequency of 40 krad/s, the additional phase is

$$360 \cdot f_T T = \frac{360^\circ}{2\pi \text{ rad}} \cdot \underbrace{40 \times 10^3 \frac{\text{rad}}{\text{s}}}_{f_T} \cdot \underbrace{\frac{2}{400 \times 10^3}}_{T} = 11.5^\circ \quad (4.1)$$

where the factor of 2 in T accounts for the two integrator delays. This value is significant and reduces the 60° phase margin to 48.5° but is still an acceptable value for stability.

SPICE simulations in time were performed on both of these circuits using macromodels and approximately 8000 steady-state data points were collected. For the SMPS, ideal component values were used. Voltage-controlled resistors with 1-mΩ ON and 10-MΩ OFF resistances were used. The elements of the passive LPF were all ideal with the exception of the ESR resistance included for the capacitor. For the compensation network, ideal components were used for the passive elements and the opamp was a voltage-controlled voltage source with an open-loop gain of 200 kV/V. Additional components were added to the opamp to provide single pole role-off with a break frequency of 35 Hz. The $\Delta\Sigma$ modulator used voltage controlled voltage sources with gains of 100 MV/V for the integrators and the comparator, the same voltage controlled resistors as the power switches, and ideal capacitors. The PWM controller used the same voltage controlled voltage source as the $\Delta\Sigma$ modulator for the comparator with a 200-kHz sawtooth wave and 0 to 5 V swing. Most of the components used can be considered ideal or nearly ideal, there are two reasons for this: ideal models allowed for faster and less data intensive simulations and second nonidealities were included where necessary.

Shown in Fig. 12 are the PSD plots of the SMPS output with PWM and $\Delta\Sigma$ modulator controllers. The clock frequency of the $\Delta\Sigma$ modulator has been adjusted to show its noise-shaping quality at different values. Also listed for these diagrams is the phase margin contributed by the $\Delta\Sigma$ modulator and, as expected, the phase contribution becomes less significant as the clock frequency is increased. The strong tones and low noise floor of the PWM controller are clearly visible in the graphs. The $\Delta\Sigma$ modulator exhibits its characteristic noise profile, with nulls occurring at integer multiples of the clock frequency. Despite the $\Delta\Sigma$ modulator's higher noise profile

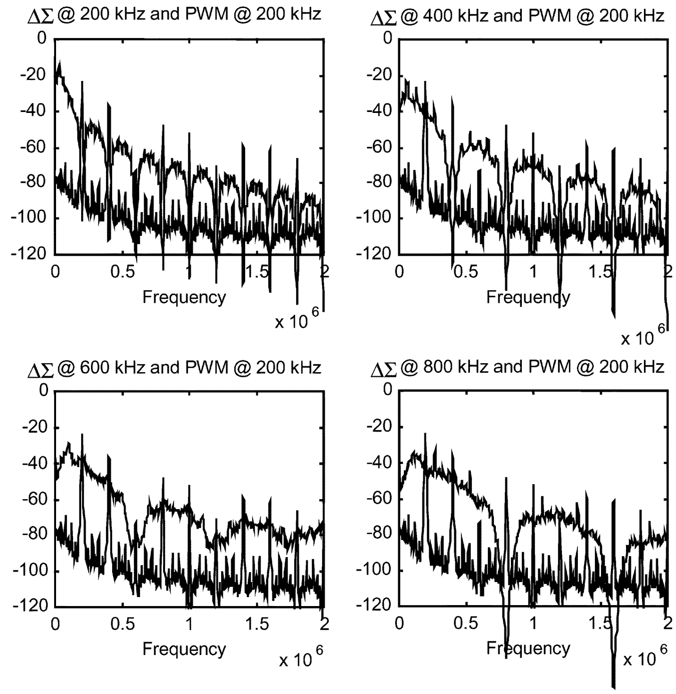

Fig. 12. PSD of $\Delta\Sigma$ modulator and PWM.

TABLE I
ADDITIVE PHASE

Clock frequency	Phase added by $\Delta\Sigma$ modulator
200 kHz	22.9 °
400 kHz	11.5 °
600 kHz	7.6 °
800 kHz	5.7 °

it is generally lower then the PWM peaks, thus produces a better signal-to-noise ratio (SNR) for an equivalent frequency region. The results for the $\Delta\Sigma$ modulator controller clocked at 200 and 400 kHz are shown in Fig. 12(a) and (b). The $\Delta\Sigma$ modulator clocked at 400 kHz shown in Fig. 12(b), has reduced the noise in the lowband region to a value equivalent to the PWMs first tone, the phase contribution has also been reduced further to 11.5°, but the efficiency of the $\Delta\Sigma$ modulator system should exceed that of the PWM. As a quantitative measure, the rms noise power for the PWM is 14.90 mW and for the $\Delta\Sigma$ modulator is 75.85 mW demonstrating the increase in overall noise in the $\Delta\Sigma$ modulator controller. Note also in the diagram that limit cycles occurring at sixths of the clock frequency have not been fully eliminated from the signal. While the limit cycles are not large enough to create stability problems, eliminating them could improve the SNR performance of the $\Delta\Sigma$ modulator; this is possible by introducing dithering or increasing the order of the $\Delta\Sigma$ modulator. For the $\Delta\Sigma$ modulator clocked at 600 kHz [Fig. 12(c)], the SNR has exceeded that of the PWM and the contribution to phase margin has been reduced to 7.6°. Using the efficiency calculations from Section III, the $\Delta\Sigma$ modulator and PWM controllers have the same efficiencies. The $\Delta\Sigma$ modulator clocked at 800 kHz [Fig. 12(d)] has noise performance and phase contributions that have improved further but the controller functions at a lesser efficiency than the PWM at this frequency. Together, these graphs, along with Table I, provide a clear picture of the design tradeoffs

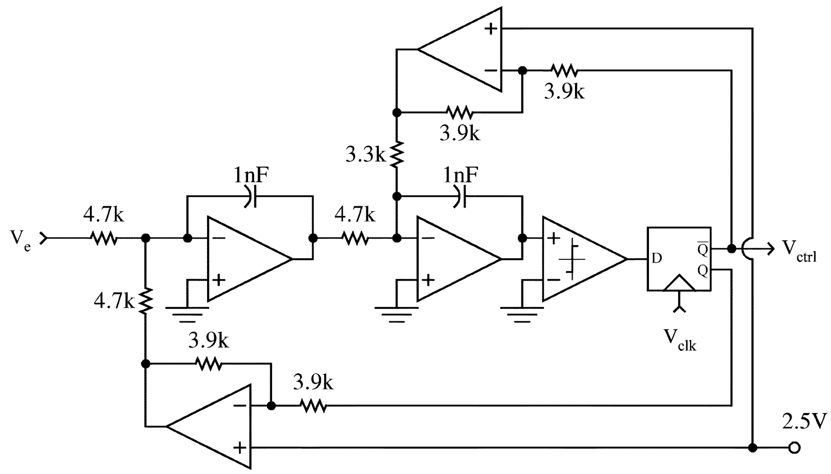

Fig. 13. Continous-time $\Delta\Sigma$ modulator.

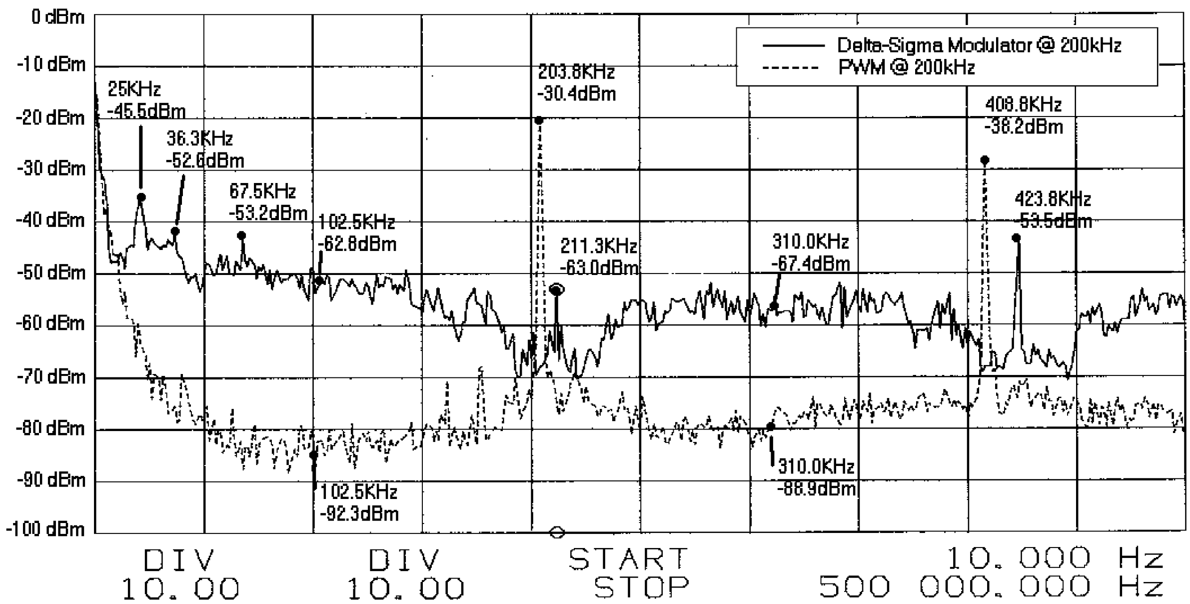

Fig. 14. Measured PSD plots for the circuit implementation of the switching power supply with the PWM (bottom) and (1-bit) $\Delta\Sigma$ modulator (top) controllers.

between the $\Delta\Sigma$ modulator and PWM controllers. For the $\Delta\Sigma$ modulator controller, stability and SNR become better as the clock frequency increases while the power transfer efficiency decreased. However, there is a clock frequency, determined by the proportionality constant, where the power transfer efficiency of the $\Delta\Sigma$ modulator is equivalent to the PWM and, assuming sufficient oversampling of the SMPS bandwidth by the $\Delta\Sigma$ modulator, the phase term contributed by the delay of the $\Delta\Sigma$ modulator is negligible. At this clock frequency, determined by k, the $\Delta\Sigma$ modulator controller creates more noise but has a larger SNR then the PWM controller.

To further validate these findings, a discrete circuit was built and tested for comparison to the simulation results. A continuous-time $\Delta\Sigma$ modulator (Fig. 13) was used in place of the switched-capacitor circuit shown earlier. The circuit was tested for the PWM clocked at 203.8 kHz and the $\Delta\Sigma$ modulator also clocked at 211.3 kHz, the resulting PSDs are shown in Fig. 14. This performance exceeds that predicted by simulation results shown in Fig. 12(a). Clock tones are visible on the $\Delta\Sigma$ modulator plot, these are due to the prototype board feedthrough. Current and voltage measurements were taken to calculate the efficiencies, these values were found to be 78% and 82% for the PWM and $\Delta\Sigma$ modulator controllers, respectively. Clearly, these efficiency numbers are not the highest reported to date, but no attempt has been made to optimize the power-supply design. Our primary objective was to examine the noise-shaping properties and compare these to a comparable PWM controller output. These results support the claim that the $\Delta\Sigma$ modulator is more efficient than the PWM at this frequency.

V. Alternative $\Delta\Sigma$ Modulator Controllers

There are three possible variations on the $\Delta\Sigma$ modulator architecture to reduce limit cycle oscillations in the SMPS. The simplest is to dither the input to the quantizer. This will increase the noise slightly but is an effective means for reducing tonal behavior [4]. The second is to increase the order of the modulator. This effectively increases the randomization at the input to the quantizer, thus reducing tones. The third approach is to employ a multibit quantizer. A multibit switching power-supply concept

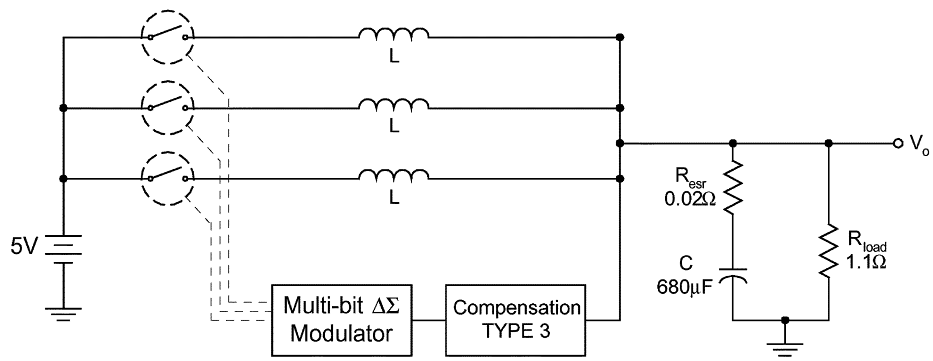

Fig. 15. Multibit switching power supply.

TABLE II
RMS POWER FOR THE PWM AND SECOND-ORDER MULTI-BIT $\Delta\Sigma$ MODULATORS SMPS CONTROLLERS

Controller Type	PWM	1-bit $\Delta\Sigma$	2-bit $\Delta\Sigma$	4-bit $\Delta\Sigma$
RMS Power (W)	14.90m	75.85m	3.75m	0.24m

is rather unusual since the PWM controller does not lend itself to this sort of topology. Designing the power supply with this in mind, a multibit implementation reduces the output noise of a switching power supply while increasing the dynamic range of the closed-loop system.

To verify that a multibit $\Delta\Sigma$ modulator controller switching power supply has better noise performance, a very simple multiswitch input voltage for a 2-bit $\Delta\Sigma$ modulator controller was created, as shown in Fig. 15. This design is impractical as a physical device because it must use multiple voltage sources. However, if considered as a "black box" design, it validates the concept of the multibit design while leaving the circuit implementation details for future development. One potential implementation strategy for this circuit involves the use of multiple inductors. The advantage of the multibit controller is that it offers smaller switching steps, in increments of 1.67 V rather than the large 0 to 5 V steps normally used. This directly effects the noise performance because the ripple voltage at the output caused by switching is scaled proportionally.

The results of the multibit switching power-supply simulations are shown in Fig. 16. These graphs show a large reduction in the rms noise at the output of the power supply with the multibit $\Delta\Sigma$ modulator controller. Table II shows the rms power of each design and provides a quantitative comparison of the ripple voltage in each design. The 1-bit $\Delta\Sigma$ modulator controller exhibits a larger rms power than the PWM controller. This is to be expected because the 1-bit $\Delta\Sigma$ has a higher noise floor than the PWM. The 2- and 4-bit $\Delta\Sigma$ modulator's, however, show a significant decrease in rms power, reflecting their greatly reduced noise profiles. A prototype was also constructed to test the multibit controller. The measured results are shown in Fig. 16(b). The output magnitude of the 2-bit system is 6 dB below the single-bit system. This is to be expected since adding a bit to the $\Delta\Sigma$ modulator will increase the dynamic range of 6 dB per bit. The efficiency is till 82% however, the rms noise has dropped to 6.3 mW. Then, the 2-bit $\Delta\Sigma$ modulator not only eliminated the tones but also dropped the overall noise power by a factor of 3.

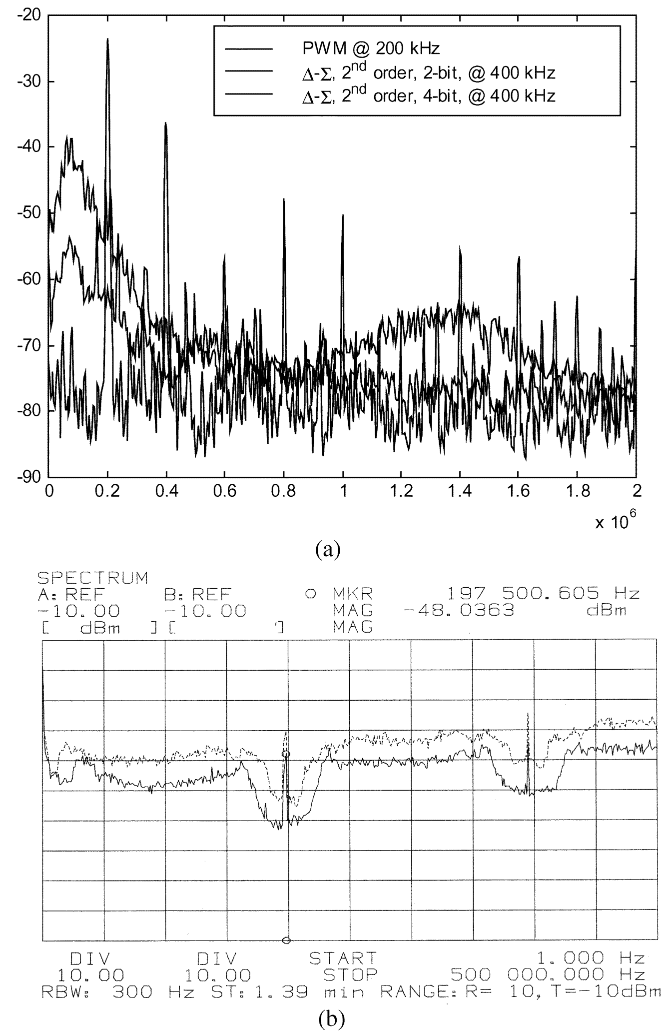

Fig. 16. (a) Simulated noise spectra of the PWM (bottom line) and $\Delta\Sigma$ modulator SMPS controllers. The middle line is the output from the 4-bit controller and the top line is the output from a 2-bit $\Delta\Sigma$ modulator controller. (b) Measured power-spectral output for a 1-bit and 2-bit $\Delta\Sigma$ SMPS controllers. The top line is for a 1-bit and the bottom is for a 2-bit controller.

VI. CONCLUSION

The SMPS is a useful circuit for converting and regulating voltage levels efficiently. The customary method of controlling a SMPS uses a PWM controller that supplies a fixed-frequency varying duty cycle. It is possible to substitute a $\Delta\Sigma$ modulator, which is a fixed period varying frequency device, for the PWM and retain the useful qualities of the SMPS. Doing this compromises a negligible amount of the closed-loop phase margin, but it gains some efficiency and better noise performance. Further, the $\Delta\Sigma$ controlling method offers other advantages such as a multibit output that may be exploited to gain even better noise performance at the cost of a greater component count.

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